Statistical Foundations for ML

TA Session 10: Maximum Likelihood Estimation & Hypothesis Testing

---

1 Part A — Maximum Likelihood Estimation

Q1. MLE for Bernoulli distribution

Problem: Let $X_1, \ldots, X_K$ be independent Bernoulli random variables with parameter p.

(a) Write the likelihood function.

(b) Obtain the log-likelihood function.

(c) Derive the maximum likelihood estimator of p.

(d) If $K = 10$ and the number of successes is 7, find the MLE.

Solution:

(a) **Likelihood Function:** Since $X_1, \ldots, X_K$ are i.i.d. Bernoulli(p) random variables, the probability mass function of a single observation is $P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}$ for $x_i \in \{0, 1\}$. The joint likelihood function $L(\underline{x}; p)$ is the product of individual PMFs:

$$L(\underline{x}; p) = \prod_{i=1}^{K} P(X_i = x_i) = \prod_{i=1}^{K} p^{x_i}(1-p)^{1-x_i} = p^{\sum_{i=1}^{K} x_i}(1-p)^{K-\sum_{i=1}^{K} x_i}$$

(b) **Log-Likelihood Function:** Taking the natural logarithm $\ln L(\underline{x}; p)$ turns products into sums, simplifying differentiation:

$$\ln L(\underline{x}; p) = \ln\left(p^{\sum_{i=1}^{K} x_i}(1-p)^{K-\sum_{i=1}^{K} x_i}\right) = \left(\sum_{i=1}^{K} x_i\right)\ln p + \left(K - \sum_{i=1}^{K} x_i\right)\ln(1-p)$$

(c) **Derivation of MLE:** Differentiate $\ln L(\underline{x}; p)$ with respect to p and set the derivative to zero:

$$\frac{d}{dp}\ln L(\underline{x}; p) = \frac{\sum_{i=1}^{K} x_i}{p} - \frac{K - \sum_{i=1}^{K} x_i}{1-p} = 0$$

Solving for the estimator $\hat{p}$:

$$\frac{\sum x_i}{\hat{p}} = \frac{K - \sum x_i}{1-\hat{p}} \implies \sum x_i(1-\hat{p}) = \hat{p}(K - \sum x_i) \implies \hat{p}K = \sum_{i=1}^{K} x_i \implies \hat{p} = \frac{1}{K}\sum_{i=1}^{K} x_i = \bar{X}$$

(d) **Numerical Calculation:** Given $K = 10$ and number of successes $\sum_{i=1}^{10} x_i = 7$:

$$\hat{p} = \frac{7}{10} = 0.7$$

Q2. MLE for Poisson distribution

Problem: Let $X_1, \ldots, X_n$ be independent Poisson random variables with unknown parameter λ.

(a) Write the likelihood function.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(b) Derive the log-likelihood function.

(c) Find the MLE of λ.

(d) If the total number of events observed over 20 days is 857, estimate the average number of events per day.

Solution:

(a) **Likelihood Function:** For i.i.d. Poisson(λ) variables, $P(X_i = x_i) = \frac{e^{-\lambda}\lambda^{x_i}}{x_i!}$. The likelihood function is:

$$L(\underline{x};\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{\sum_{i=1}^{n} x_i}}{\prod_{i=1}^{n} x_i!}$$

(b) **Log-Likelihood Function:** Taking the natural logarithm:

$$\ln L(\underline{x};\lambda) = \ln(e^{-n\lambda}) + \ln(\lambda^{\sum x_i}) - \ln\left(\prod_{i=1}^{n} x_i!\right) = -n\lambda + \left(\sum_{i=1}^{n} x_i\right)\ln\lambda - \sum_{i=1}^{n}\ln(x_i!)$$

(c) **Finding the MLE:** Differentiating w.r.t. λ and setting to zero:

$$\frac{d}{d\lambda}\ln L(\underline{x};\lambda) = -n + \frac{\sum_{i=1}^{n} x_i}{\hat{\lambda}} = 0 \implies \hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

(d) **Numerical Estimation:** Given total events over $n = 20$ days is $\sum x_i = 857$, the estimated average daily event rate is:

$$\hat{\lambda} = \frac{857}{20} = 42.85 \text{ events/day}$$

Q3. MLE and likelihood

Problem: Which of the following statements are true?

(A) The MLE is the parameter value that maximizes the likelihood of the observed data.

(B) The likelihood function is a probability distribution over the possible values of the parameter.

(C) Maximizing the log-likelihood gives the same MLE as maximizing the likelihood.

(D) The MLE must always be an unbiased estimator.

Solution:

Correct Options: Statements (A) and (C) are true.

- **(A) True:** Maximum Likelihood Estimation is formally defined as searching over the parameter space to find $\hat{\theta}$ that maximizes $L(\theta|\underline{x})$.
- **(B) False:** The likelihood function integrates or sums to 1 with respect to the data space (for a fixed parameter), but it is not a valid probability density function over the parameter space because its integral over θ does not necessarily equal 1.
- **(C) True:** Because the natural logarithm $\ln(x)$ is a strictly monotonically increasing function, any argument θ that maximizes $L(\theta)$ will also maximize $\ln L(\theta)$.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

- **(D) False:** MLEs possess asymptotic properties (like asymptotic unbiasedness and consistency), but they can be biased in small/finite samples (e.g., the MLE of variance in a normal distribution divides by n instead of $n-1$, making it biased).

2 Part B — Hypothesis Testing

Q4. Understanding p-values [MCQ/MSQ]

Problem: Which of the following are true about p-values?

(A) The p-value gives the probability of making a Type I error.

(B) The p-value is a measure of how extreme the observed data are under the null hypothesis.

(C) A p-value below the significance level allows us to conclude with certainty that the null hypothesis is false.

(D) If the null hypothesis is true, then the p-value will always be larger than the significance level.

Solution:

Correct Option: Statement (B) is true.

- **(A) False:** The probability of making a Type I error (α) is a predetermined significance threshold set by the experimenter before observing data, whereas the p-value is computed directly from the observed sample data.
- **(B) True:** The p-value is defined as the probability of obtaining a test statistic as extreme as, or more extreme than, the observed value, assuming that the null hypothesis H_0 is true.
- **(C) False:** Hypothesis testing is probabilistic; rejecting H_0 means there is strong statistical evidence against it, but it never grants absolute "certainty."
- **(D) False:** When H_0 is true, the p-value is distributed uniformly on the interval $[0, 1]$. Thus, it falls below α (e.g., 0.05) exactly 5% of the time purely by random chance (Type I error).

Q5. Type I and Type II errors [MCQ]

Problem: The rejection of a false null hypothesis is a:

(i) Type I error

(ii) Type II error

Which of the following is true?

(A) (i)

(B) (ii)

(C) Both (i) and (ii)

(D) Neither (i) nor (ii)

Solution:

Correct Option: (D) Neither (i) nor (ii).

- **Type I Error:** Rejecting a true null hypothesis H_0 (False Positive).

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

- **Type II Error:** Failing to reject a false null hypothesis H_0 (False Negative).
- **Correct Decision:** Rejecting a false null hypothesis means our test successfully detected an effect. This is a correct statistical decision, and its probability is known as the Power of the Test $(1 - \beta)$. Therefore, it is neither (i) nor (ii).

Q6. Type I and Type II errors [MCQ]

Problem: In a hypothesis-testing problem, which of the following statements are **not** true?

(A) The probability of a Type I error cannot be higher than the probability of a Type II error.

(B) Type II error occurs if the test fails to reject the null hypothesis when the null hypothesis is actually false.

(C) Type I error occurs if the test rejects the null hypothesis when the null hypothesis is actually false.

(D) The sum of the probability of Type I error and the probability of Type II error should be 1.

Solution:
Not True Statements: (A), (C), and (D).

- **(A) Not true:** α and β are independent error rates governed by sample size and critical region placement; one can easily be higher than the other.
- **(B) True:** This is the textbook definition of a Type II error (β).
- **(C) Not true:** Type I error occurs when rejecting a *true* null hypothesis (H_0 is true, but we falsely reject it). Rejecting a false H_0 is correct.
- **(D) Not true:** $\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$ and $\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$. Because they are conditioned on mutually exclusive states of nature, $\alpha + \beta \neq 1$.

Q7. Hypothesis formulation [Short answer]

Problem: For each of the following claims, formulate the null and alternative hypotheses:

(a) A population has mean $\mu = 8$.

(b) A population has mean $\mu > 8$.

Identify which hypotheses are simple and which are composite. **Solution:**
Formulations & Classifications:

(a) **Claim ($\mu = 8$):** $H_0 : \mu = 8$ versus $H_1 : \mu \neq 8$.
Classification: H_0 specifies an exact single value $(\mu = 8) \implies$ Simple Hypothesis. H_1 specifies a range $(\mu \neq 8) \implies$ Composite Hypothesis.

(b) **Claim ($\mu > 8$):** $H_0 : \mu \leq 8$ versus $H_1 : \mu > 8$.
Classification: H_0 covers a range $(\mu \leq 8)$ and H_1 covers a range $(\mu > 8) \implies$ Both are Composite Hypotheses.

Q8. Testing a population mean with known variance [Numerical]

Problem: Suppose X is normally distributed with mean μ and known standard deviation $\sigma = 2$. We wish to test:

$$H_0 : \mu = 8 \quad \text{versus} \quad H_1 : \mu \neq 8$$

Five independent samples are observed, and the sample mean is $\bar{X} = 9.5$.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(a) Find the test statistic.

(b) Determine the rejection region for a 5% significance level.

(c) Should H_0 be rejected at the 5% level?

(d) Repeat the decision for a 10% significance level.

(e) Explain why changing the significance level can change the decision.

Solution:

(a) **Test Statistic:** Given $\mu_0 = 8$, $\sigma = 2$, $n = 5$, and $\bar{X} = 9.5$:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{9.5 - 8}{2/\sqrt{5}} = \frac{1.5}{2/2.23607} = \frac{1.5}{0.89443} \approx 1.677$$

(b) **Rejection Region ($\alpha = 0.05$):** For a two-tailed test, critical values are $\pm z_{0.025} = \pm 1.96$. Rejection region is $|Z| > 1.96$.

(c) **Decision at 5% Level:** Since $|1.677| \not> 1.96$, fail to reject H_0 at $\alpha = 0.05$.

(d) **Decision at 10% Level ($\alpha = 0.10$):** Critical value is $z_{0.05} = 1.645$. Rejection region is $|Z| > 1.645$. Since $|1.677| > 1.645$, reject H_0 at $\alpha = 0.10$.

(e) **Why Decision Changes:** Increasing α expands the critical rejection region, lowering the evidence barrier required to reject H_0. When a test statistic falls near the boundary, this expansion can flip the test outcome.

Q9. Hypothesis testing using p-values [Numerical]

Problem: Consider the same setting as Q8 ($H_0 : \mu = 8$ versus $H_1 : \mu \neq 8$ with known $\sigma = 2$ and sample size $n = 5$). Calculate the p-value and determine whether H_0 should be rejected at the 5% significance level for each of the following sample means:

(a) $\bar{X} = 8.5$

(b) $\bar{X} = 9.5$

(c) $\bar{X} = 11.5$

Explain the relationship between the p-value and the strength of evidence against H_0. **Solution:**

(a) For $\bar{X} = 8.5$: $Z = \frac{8.5-8}{2/\sqrt{5}} = 0.559$.
p-value $= 2P(Z > 0.559) = 2(0.288) = 0.576 > 0.05 \implies$ Fail to reject H_0.

(b) For $\bar{X} = 9.5$: $Z = 1.677$.
p-value $= 2P(Z > 1.677) = 2(0.0467) = 0.0934 > 0.05 \implies$ Fail to reject H_0.

(c) For $\bar{X} = 11.5$: $Z = \frac{11.5-8}{2/\sqrt{5}} = 3.91$.
p-value $= 2P(Z > 3.91) \approx 0.00009 < 0.05 \implies$ Reject H_0.

Relationship: The p-value measures the exact data extremity under H_0. Smaller p-values indicate greater discrepancy between sample data and H_0, representing stronger evidence against H_0.

Q10. Rejection region and significance level [Short answer]

Problem:

(a) Define the rejection region of a hypothesis test.

(b) Explain how the rejection region is determined by the significance level α.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(c) What happens to the rejection region when α is increased?

(d) Explain the effect of increasing α on the probability of a Type I error and the probability of a Type II error.

Solution:

(a) The rejection (or critical) region is the set of all possible values of the test statistic that lead to the rejection of the null hypothesis H_0.

(b) The significance level α dictates the total probability allocated to the critical region under the assumption that H_0 is true: $P(\text{Test Statistic} \in \text{Rejection Region} \mid H_0) = \alpha$.

(c) Increasing α expands the rejection region (moves the critical boundaries inward), making it easier to reject H_0.

(d) Increasing α directly increases the probability of a Type I error (α), while simultaneously decreasing the probability of a Type II error (β), demonstrating the fundamental trade-off between the two error types.

Q11. Power of a hypothesis test [Numerical]

Problem: A test is performed for $H_0 : \mu = 8$ versus $H_1 : \mu \neq 8$. Assume the population is normal with known standard deviation $\sigma = 2$, and the sample size is $n = 5$.

(a) Write the rejection region for a 5% significance level.

(b) Suppose the true mean is $\mu = 10$. Express the probability of rejecting H_0 in terms of a standard normal random variable.

(c) What is the power of the test when $\mu = 10$?

(d) Explain qualitatively how increasing the sample size affects the power of the test.

Solution:

(a) **Rejection Region ($\alpha = 0.05$):** For a two-tailed test, $z_{0.025} = 1.96$. The rejection region in terms of $\bar{X}$ is:

$$\left|\frac{\bar{X} - 8}{2/\sqrt{5}}\right| > 1.96 \implies \bar{X} < 8 - 1.96\left(\frac{2}{\sqrt{5}}\right) \text{ or } \bar{X} > 8 + 1.96\left(\frac{2}{\sqrt{5}}\right)$$

$$\implies \bar{X} < 8 - 1.7518 = 6.2482 \text{ or } \bar{X} > 8 + 1.7518 = 9.7518$$

(b) **Probability of Rejection (Power when $\mu = 10$):** The power is the probability of falling into the rejection region given that $\mu = 10$:

$$\text{Power} = P(\bar{X} < 6.2482 \mid \mu = 10) + P(\bar{X} > 9.7518 \mid \mu = 10)$$

Standardizing $\bar{X}$ under true mean $\mu = 10$ where $\text{SE} = \frac{2}{\sqrt{5}} \approx 0.89443$:

$$= P\left(Z < \frac{6.2482 - 10}{0.89443}\right) + P\left(Z > \frac{9.7518 - 10}{0.89443}\right) = P(Z < -4.195) + P(Z > -0.2764)$$

(c) **Power Calculation:** Evaluating using standard normal probabilities:

$$\text{Power} = P(Z < -4.195) + P(Z > -0.2764) \approx 0.000013 + 0.60888 = 0.609 \ (60.9\%)$$

(d) **Effect of Sample Size:** Increasing sample size n reduces the standard error ($\sigma/\sqrt{n}$), which shifts the sampling distribution under H_1 further away from the critical thresholds, thereby increasing the power of the test.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

3 Part C — Homework

Q12. Bernoulli hypothesis test [Numerical]

Problem: A random sample $X_1, \ldots, X_6$ of size 6 is taken from a Bernoulli distribution with parameter θ. The null hypothesis $H_0 : \theta = \frac{1}{2}$ is to be tested against the alternative $H_1 : \theta > \frac{1}{2}$, based on the statistic $Y = \sum_{i=1}^{6} X_i$. If the observed value of Y is 4, find the p-value of the test.

(A) $\frac{21}{32}$

(B) $\frac{9}{64}$

(C) $\frac{11}{32}$

(D) $\frac{7}{64}$

Solution:

Correct Option: (C) $\frac{11}{32}$

Detailed Working Steps: Under $H_0 : \theta = 1/2$, the test statistic $Y = \sum_{i=1}^{6} X_i \sim \text{Binomial}(6, 1/2)$. For a right-tailed alternative $H_1 : \theta > 1/2$, the p-value given observed $Y = 4$ is:

$$\text{p-value} = P(Y \geq 4 \mid \theta = 1/2) = \sum_{k=4}^{6} \binom{6}{k} \left(\frac{1}{2}\right)^6 = \frac{1}{64}\left[\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right]$$

Evaluating the binomial coefficients:

$$\binom{6}{4} = 15, \quad \binom{6}{5} = 6, \quad \binom{6}{6} = 1 \implies \text{Sum} = 15 + 6 + 1 = 22$$

$$\text{p-value} = \frac{22}{64} = \frac{11}{32}$$

Q13. Conceptual question [Short answer]

Problem: A hypothesis test is performed at significance level $\alpha = 0.05$, and the resulting p-value is 0.03.

(a) What is the decision of the test?

(b) Can we conclude that the null hypothesis is false with probability 0.97?

(c) Explain the difference between the significance level and the p-value.

Solution:

(a) Since p-value$(0.03) < \alpha(0.05)$, we reject the null hypothesis H_0.

(b) No. In frequentist hypothesis testing, parameters and hypotheses are treated as fixed, non-random population truths (either true or false). We cannot assign a probability value (like 0.97) to H_0 being false.

(c) Significance level (α) is the pre-chosen threshold for tolerable Type I error; p-value is computed post-experiment from sample data to measure the strength of evidence against H_0.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.